

MINERALNYE VODY, Russian Federation

WMO: 370540

Lat: 44.225N

Lon: 43.082E

Elev: 1054

StdP: 14.14

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.1	8.3	-4.0	4.6	1.3	3.4	6.8	8.7	24.9	28.0	22.4	25.7	8.1	100	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.1	94.5	69.7	91.3	68.7	87.9	68.0	72.5	87.9	71.0	85.6	69.7	83.8	10.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.8	106.3	79.1	66.3	100.7	77.1	64.8	95.4	75.5	36.9	88.0	35.6	86.0	34.4	83.7	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.9	20.2	18.0	-5.3	98.7	9.4	3.9	-12.0	101.5	-17.5	103.8	-22.8	105.9	-29.6	108.8		
(5)			-5.6	75.5	8.9	1.9	-12.0	76.9	-17.2	78.0	-22.1	79.1	-28.6	80.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	28.0	30.2	39.7	50.7	60.7	68.8	74.0	73.7	64.0	51.9	40.1	31.3		
	DBStd	17.91	8.88	9.85	7.18	6.98	6.02	5.29	4.64	5.31	6.66	7.93	7.46	8.56		
	HDD50	2601	682	556	328	74	5	0	0	0	2	72	303	579		
	HDD65	5815	1147	976	786	429	155	24	1	5	94	407	747	1044		
	CDD50	3045	0	0	7	96	338	565	744	735	422	132	6	0		
	CDD65	786	0	0	0	1	23	138	281	275	65	3	0	0		
	CDH74	8691	0	0	3	53	374	1471	3087	2939	692	72	0	0		
	CDH80	3583	0	0	0	8	69	513	1405	1361	209	18	0	0		
(14) Wind	WSAvg	7.9	8.0	8.6	9.1	8.9	7.7	7.3	7.2	7.4	7.4	7.8	7.9	7.8		
(15) Precipitation	PrecAvg	21.6	0.8	0.8	1.3	2.2	2.7	3.7	2.6	1.8	1.9	1.6	1.1	1.1		
	PrecMax	27.0	1.6	1.5	2.7	6.0	4.4	9.9	7.2	4.3	4.7	3.5	2.6	2.2		
	PrecMin	13.6	0.2	0.3	0.4	0.9	0.7	1.1	0.6	0.3	0.4	0.2	0.2	0.2		
	PrecStd	3.4	0.4	0.3	0.6	1.1	1.0	1.9	1.4	1.1	1.0	0.9	0.6	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.8	59.2	71.3	79.8	85.7	93.4	98.4	98.4	91.8	82.5	66.4	55.5		
		MCWB	43.5	48.0	54.8	59.6	65.7	68.0	71.5	70.5	68.3	63.2	53.7	46.9		
	2%	DB	45.0	51.9	62.8	73.8	81.9	89.4	94.6	94.7	85.6	74.0	59.3	48.6		
		MCWB	39.9	44.5	50.1	57.9	64.1	67.9	70.1	69.9	65.8	60.4	51.0	43.7		
	5%	DB	40.8	45.9	57.2	69.7	78.5	85.8	91.2	91.3	81.1	69.4	54.0	44.7		
		MCWB	37.3	40.6	47.5	55.8	63.2	67.6	69.0	68.7	64.2	58.5	48.4	41.2		
	10%	DB	37.6	41.2	51.9	64.8	75.0	82.4	87.9	87.8	77.4	64.4	50.1	41.5		
		MCWB	35.2	37.7	45.0	53.7	62.0	66.4	68.2	67.8	63.1	56.3	46.5	39.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.4	49.1	55.3	62.1	68.4	73.3	74.6	74.1	71.2	63.9	55.2	47.6		
		MCDB	50.3	58.0	68.0	75.5	81.7	85.8	91.1	90.9	87.3	77.0	63.5	53.1		
	2%	WB	40.4	44.3	51.1	59.0	65.9	70.7	72.6	72.2	67.7	61.4	51.9	44.0		
		MCDB	44.5	50.9	60.8	71.3	78.1	83.7	88.2	88.0	79.8	71.7	58.5	47.7		
	5%	WB	37.6	40.9	48.3	56.9	64.2	69.0	71.1	70.8	65.9	59.5	49.2	41.6		
		MCDB	40.1	45.0	55.9	67.9	75.5	81.7	85.8	85.8	77.7	67.3	53.4	44.2		
	10%	WB	35.2	37.9	45.5	54.6	62.4	67.4	69.8	69.5	64.2	57.2	46.7	39.4		
		MCDB	37.0	40.5	51.3	63.6	73.1	79.4	84.1	83.6	75.0	63.6	49.8	41.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	15.1	16.8	20.9	21.6	22.7	24.1	23.9	21.8	17.9	14.6	13.0		
		MCDBR	19.5	23.7	29.8	30.6	28.9	29.2	30.4	29.8	29.4	28.3	24.5	19.6		
		MCWBR	14.7	16.3	18.2	16.2	14.2	12.0	11.1	10.8	12.7	15.6	16.4	14.6		
		MCDBR	18.0	21.9	27.4	27.1	25.5	25.1	26.5	26.2	24.8	24.7	22.6	17.9		
	5% WB	MCWBR	13.9	15.4	17.4	15.1	13.3	11.9	11.0	11.0	12.0	14.1	15.5	13.6		
(40) Clear-Sky Solar Irradiance	taub		0.327	0.362	0.407	0.461	0.429	0.435	0.467	0.469	0.426	0.385	0.345	0.328		
	taud		2.317	2.232	2.148	2.035	2.215	2.244	2.165	2.157	2.247	2.319	2.372	2.322		
	Ebn at Noon		247	257	259	254	268	266	255	250	251	246	239	233		
	Edh at Noon		26	34	42	51	44	43	46	45	38	31	25	24		
(44) All-Sky Solar Radiation	RadAvg		441	706	950	1411	1741	1943	1950	1759	1306	807	487	364		
	RadStd		79	138	105	161	181	117	128	109	144	110	70	49		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+0.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+243	N/A

Nomenclature: See separate page